

ZAID AHMED

AI-ML Engineer | Python Developer

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SUMMARY

AI/ML Engineer with 3+ years of experience in Python and machine learning, building computer vision, speech and LLM systems end to end, from data and models to APIs and GPU inference with NVIDIA Triton, DeepStream, vLLM and TensorRT. Open-source contributor passionate about advancing AI technology and delivering scalable, high-impact solutions. Recognized for technical expertise, reliability, and driving projects end-to-end from concept to production.

EXPERIENCE

AI-ML Engineer/Python Developer

Inseyab Consulting and Information solutions / Trade foresight

07/2025 - Present | NASTP, Sharah e Faisal, Karachi

An AI-focused firm building intelligent solutions that enhance business efficiency and strategic decision-making.

- Built a single-GPU inference platform on NVIDIA Triton serving vision, LLM/VLM (Gemma 4), speech, embedding and OCR models behind an OpenAI-compatible FastAPI gateway over HTTP and WebSocket.
- Doubled YOLO segmentation throughput to 121 req/s with TensorRT and cut peak model VRAM by 76% by reworking ONNX Runtime memory management on a constrained GPU budget.
- Worked on the Voice Intelligence Platform: Genesys call transcription, sentiment, compliance filtering, semantic search and a live English/Arabic voice agent.
- Led end-to-end development of AI-driven solutions, covering backend architecture, model integration, and API design.
- Designed and developed RESTful APIs using Python frameworks (e.g., FastAPI, Flask) and robust schemas with optimized queries for PostgreSQL and MySQL.
- Built and optimized data pipelines for data preprocessing, feature engineering, and efficient model inference workflows.
- Containerized backend services using Docker and managed cloud deployments on AWS and Azure environments.

AI-ML Engineer

Contract Base / Freelance

01/2023 - Present

AI-ML Engineer

Tesseract Innovations (Pvt) Ltd

06/2024 - 06/2025 | Gulshan-e-Iqbal, Karachi

A tech firm focused on innovative AI solutions.

- Designed and deployed end-to-end machine learning models, from data pre-processing to production-ready solutions.
- Applied computer vision techniques for object detection, segmentation, and pose estimation across various use cases.
- Led API integration for seamless model-to-product transitions in real-time applications.
- Performed in-depth data analysis, feature engineering, and hyperparameter tuning to enhance model performance.
- Conducted model evaluation and implemented continuous improvement strategies for deployed systems.
- Implemented Agentic AI concepts to build more autonomous and decision-capable AI systems.

PROJECTS

Triton Inference Platform

Unified GPU inference stack for vision, language and speech models

- One Triton server with onnxruntime, vLLM and Python backends: YOLO, InsightFace, Gemma 4 (text + vision), PaddleOCR-VL, Whisper, TTS and embeddings.
- OpenAI-compatible gateway with per-model FIFO queues, access keys, per-origin limits and camera stream admission.
- Operator console with live GPU/VRAM telemetry, runtime model load/unload and VRAM reservation budgeting.
- Measured 121 req/s YOLO-seg with TensorRT (+104%) and 93 ms VLM time-to-first-token; 27 defects found and fixed across live audits.
- DeepStream multi-camera ingestion and an optional Kafka plane for asynchronous jobs.

Voice Intelligence platform

AI-Driven Automated Customer Support System

- Built a real-time conversational AI system with Whisper ASR and natural TTS.
- Enhanced transcription using diarization, normalization, and noise reduction.
- Designed agentic AI workflows with context-aware NLP and smart escalation.
- Implemented vector-based semantic search for accurate information retrieval.
- Utilized Apache Kafka for scalable, low-latency streaming and queue management.

Live Stream Analysis & Tracking

Real-time multimodal video understanding

- Object detection and tracking on live camera feeds at ~30 FPS.
- Gemma 4 vision-language model summarises a frame montage every few seconds into plain-English scene reports.
- Served through NVIDIA Triton for high-throughput inference.

Web3 Trading Chatbot

A crypto trading guide with action-ready agents inspired by Chatoshi platform

- AI-powered crypto assistant that understands natural-language commands to Buy / Sell / Transfer crypto and copy trades from other wallets.
- Action-ready agents for market updates, trade execution and strategy suggestions across Ethereum, Polygon and Solana.
- Voice command support for hands-free interaction.

EXPERIENCE

Python Developer (Internship)

Codes Soft

10/2023 - 12/2023 | Gulshan-e-Iqbal, Karachi

A technology company specializing in ERP software development.

- Contributed to feature development and customized ERP applications.
- Collaborated on real-world projects and enhanced Python development skills.

EDUCATION

Bachelors of Science in Software Engineering

Sir Syed University of Engineering & Technology

08/2020 - 06/2024 | Karachi

Artificial Intelligence and Data Sciences

Saylani Mass IT (SMIT)

01/2022 - 12/2023 | Karachi

SKILLS

AI EXPERTISE

Python

Generative AI (GenAI)

Agentic AI

RAG

Computer Vision

OCR

Vision-Language Models

Speech To Text (Whisper)

Text To Speech

Tensorflow

PyTorch

Feature Engineering

Data Pre-processing

Data Annotation

Image Analysis

Video Analysis

Voice Analysis

Vector Databases

Scikit-learn

Fine-tuning Models

Model Deployment

LangChain

LangGraph

ENGINEERING & INFRASTRUCTURE

Backend Development (FastAPI/Django/Flask)

NVIDIA Triton

vLLM

TensorRT

ONNX Runtime

DeepStream

Roboflow

CI/CD

Claude Code/API

OpenAI API

Docker

Kafka

Redis

WebSockets

PostgreSQL

MySQL

CLOUD PLATFORMS & TOOLS

AWS

Azure

GCP

Vast.ai

RunPod

LANGUAGES

English

Proficient

●●●●●

Urdu

Native

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PROJECTS

Twitter/X Engagement Agent

A custom-built Twitter/X bot for the official Game5Ball page.

- Automated, context-aware engagement through replies, likes and retweets to boost visibility.
- AI-generated posts, including narrated news videos with subtitles, with semantic de-duplication so content never repeats.

MeasureMates

An AI project focused on measuring and analyzing cow postures.

- Developed a 2D pose estimation & segmentation model of livestock animals (cows).
- Managed every step from data collection and annotating process to model training and deployment.
- Backend development for seamless integration via API's.

3D Teeth Segmentation

Per-tooth segmentation of 3D intraoral scans

- MeshSegNet-based pipeline with graph-cut refinement for upper and lower jaws, exported as colour-coded .glb models.

PUBLICATIONS

Open Source Datasets

Kaggle

04/2024 | kaggle.com/datasets/zaidworks0508

Led the gathering, labelling, analysis and model training process for cow pose, segmentation and breed datasets, with 1,300+ downloads.

CERTIFICATIONS

✓ AI and Data Science From SMIT

✓ Building With Claude API By Anthropic

✓ Introduction To Model Context Protocol By Anthropic

AWARDS

🏆 Sir Syed Technovative Ideathon Winner

Winner of Sir Syed Technovative Ideathon 2024.

🏆 Devathon SUMIT 1.0 Laptop Winner

Won the Laptop in 'AI & DS' Category at Devathon SUMIT 1.0.

🏆 2nd Position in AI Project Exhibition

Achieved 2nd Position in University AI Project Exhibition 2022.